

RESEARCH PROFILE

PhD researcher working primarily on open quantum systems and non-equilibrium quantum dynamics. Research interests include quantum many-body dynamics, relaxation and transport, dissipation and decoherence, non-Hermitian physics, statistical mechanics, quantum information, quantum simulation, and connections with machine learning.

CURRENT AFFILIATION

Current **PhD Candidate**
Institute of Physics, Chinese Academy of Sciences · Beijing, China

RESEARCH INTERESTS

- Non-equilibrium quantum dynamics
- Quantum many-body physics
- Statistical mechanics
- Open quantum systems
- Non-Hermitian physics
- Quantum information and simulation

PUBLICATIONS AND PREPRINTS

- 2026 **[Geometry-adaptive formulation of non-Bloch bands in arbitrary dimensions and spectral instability ↗](#)**
Ze-Yu Xing, Yuncheng Xiong, and Haiping Hu
Communications Physics 9, 127 (2026)
Ze-Yu Xing and Yuncheng Xiong contributed equally.
- 2025 **[Universal spreading dynamics in quasiperiodic non-Hermitian systems ↗](#)**
Ze-Yu Xing, Shu Chen, and Haiping Hu
Physical Review B 111, L180203 (2025)
- 2024 **[Non-Hermitian skin effect in arbitrary dimensions: non-Bloch band theory and classification ↗](#)**
Yuncheng Xiong, **Ze-Yu Xing**, and Haiping Hu
Preprint, *arXiv:2407.01296* (2024)

ACADEMIC PROFILES

Google Scholar [Profile ↗](#)

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